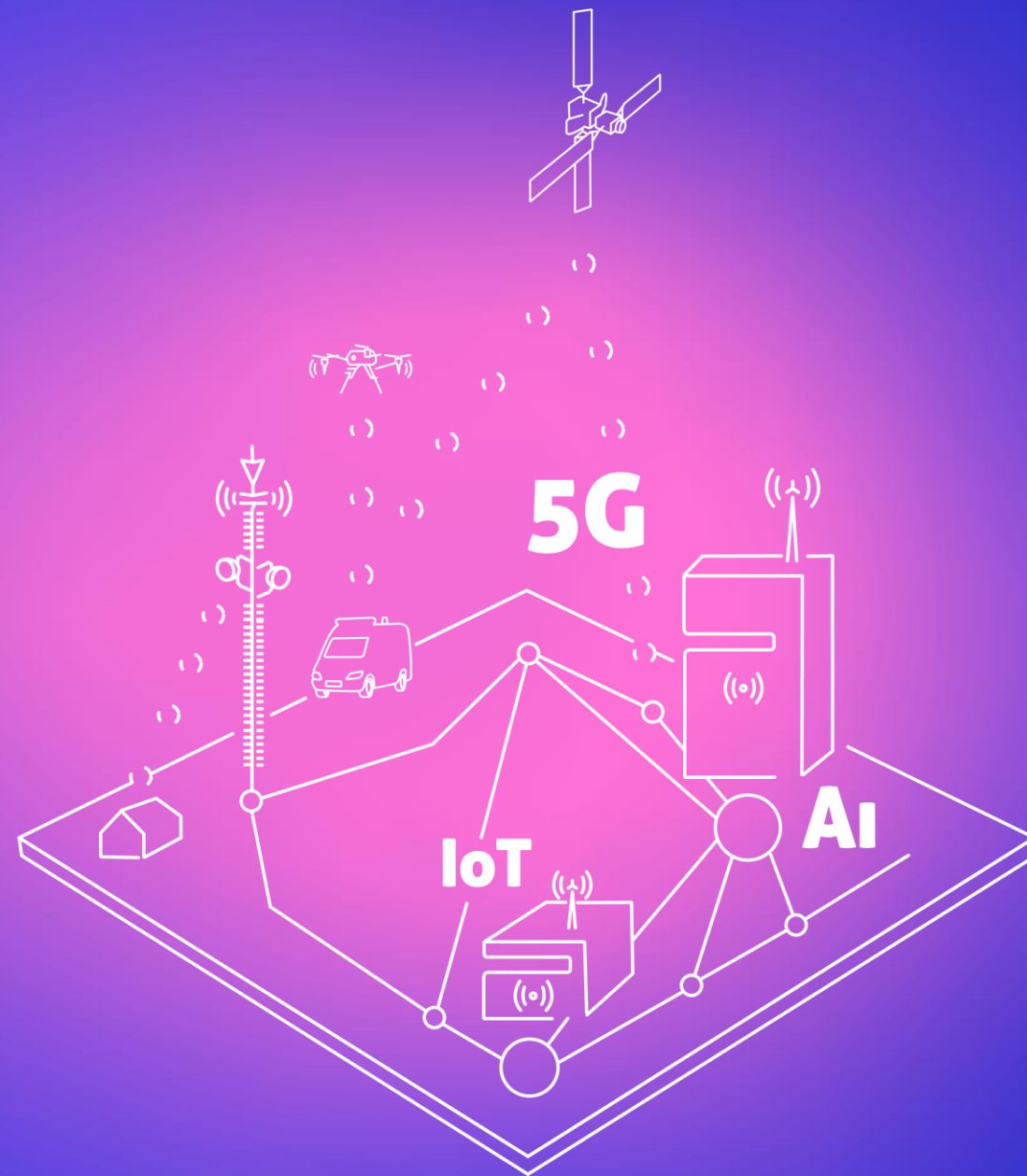




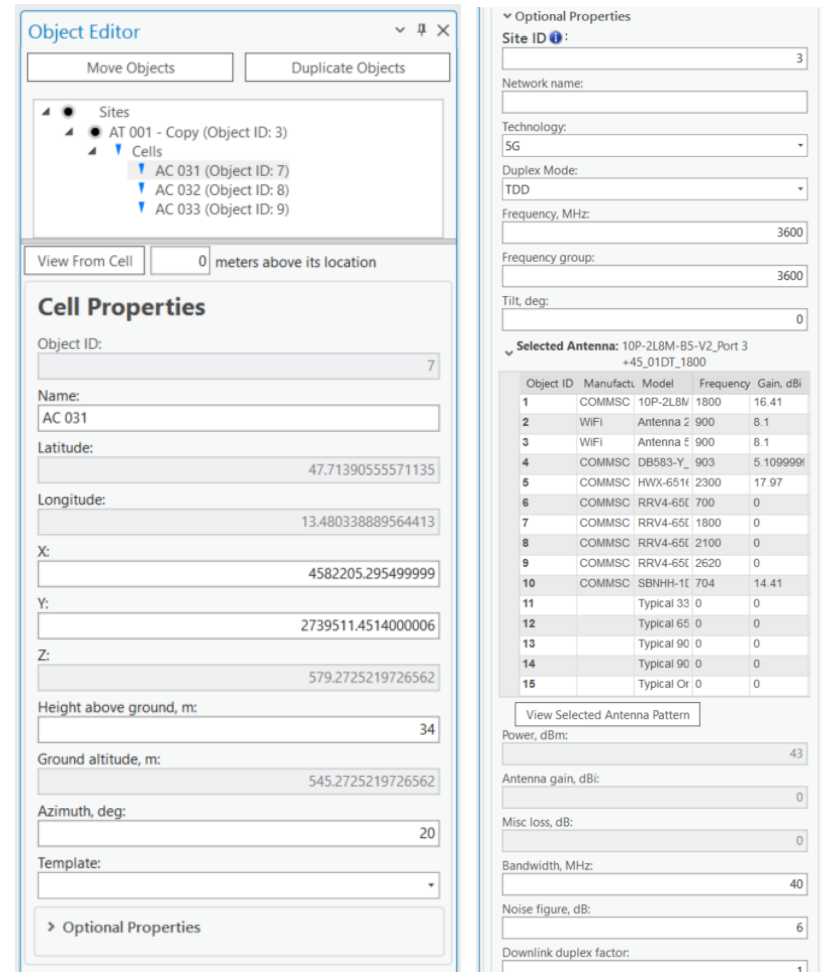
4. Cell prediction

www.cellular-expert.com



Cell structure

- Physical parameters
 - Coordinates
 - Height
 - Azimuth
 - ...
- Logical parameters
 - Power
 - Bandwidth
 - Frequency
 - ...



The screenshot displays the Cellular Expert software interface. The 'Object Editor' panel on the left shows a tree view with 'Sites' containing 'AT 001 - Copy (Object ID: 3)' and 'Cells' containing 'AC 031 (Object ID: 7)', 'AC 032 (Object ID: 8)', and 'AC 033 (Object ID: 9)'. Below this is a 'View From Cell' dropdown set to '0' meters above its location. The 'Cell Properties' panel for 'AC 031' includes fields for Object ID, Name, Latitude, Longitude, X, Y, Z, Height above ground, Ground altitude, Azimuth, and Template. The 'Optional Properties' panel on the right includes fields for Site ID, Network name, Technology, Duplex Mode, Frequency, Frequency group, Tilt, and a table for 'Selected Antenna' properties.

Object ID	Manufact.	Model	Frequency	Gain, dBi
1	COMMSC	10P-2L8M	1800	16.41
2	WiFi	Antenna 2	900	8.1
3	WiFi	Antenna 2	900	8.1
4	COMMSC	DB583-Y	903	5.109999
5	COMMSC	HWX-6511	2300	17.97
6	COMMSC	RRV4-651	700	0
7	COMMSC	RRV4-651	1800	0
8	COMMSC	RRV4-651	2100	0
9	COMMSC	RRV4-651	2620	0
10	COMMSC	SBNHH-11	704	14.41
11		Typical 33	0	0
12		Typical 65	0	0
13		Typical 90	0	0
14		Typical 90	0	0
15		Typical Or	0	0

- Projected coordinate system:
 - X
 - Y
- Geographic coordinate system in meters:
 - Longitude
 - Latitude
- Z – total cell height above sea level.

Latitude:	47.7072180136647
Longitude:	13.609511659472973
X:	4591926.4914
Y:	2739233.6751000006
Z:	486.9194641113281

▼ Spatial Reference

Projected Coordinate System	ETRS 1989 LAEA
Projection	Lambert Azimuthal Equal Area
WKID	3035
Authority	EPSG
Linear Unit	Meters (1.0)
False Easting	4321000.0
False Northing	3210000.0
Central Meridian	10.0

- Unique parameter in the project.
- Best server – the same.

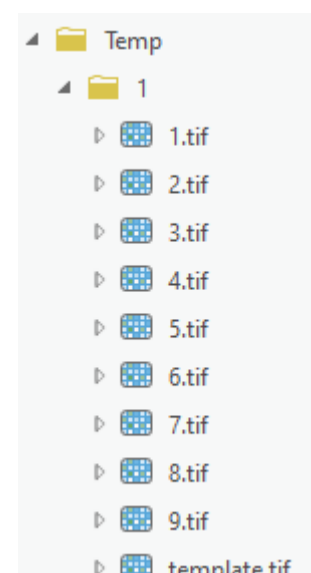
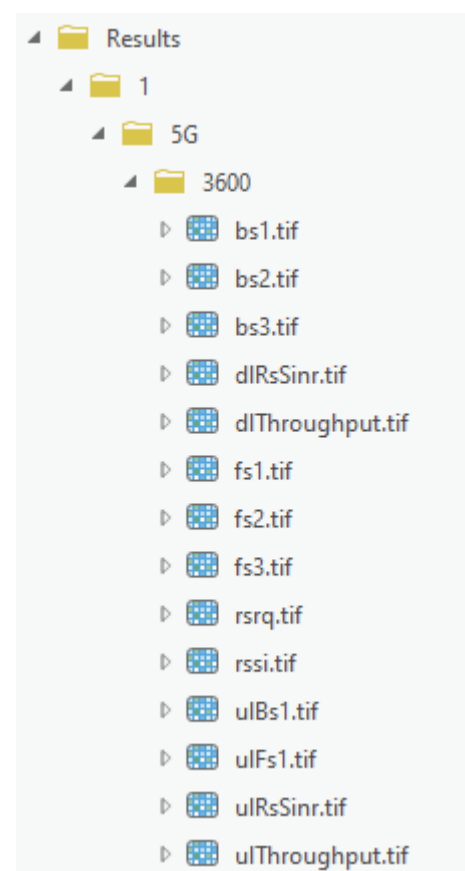
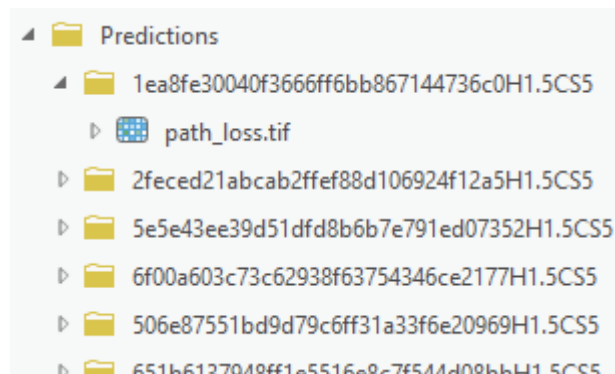
Name:	
	AC 023
Latitude:	

General cells parameter

Value	CE Field	Units	Example	Description
Latitude	latitude	Meters	49.9993	Y point coordinate in Decimal degrees and in WGS 1984 geographical coordinate system.
Longitude	longitude	Meters	33.6573	X point coordinate in Decimal degrees and in WGS 1984 geographical coordinate system.
Cell identification	cell_name	[text]	5G cell XXYY	Represents cell identification, usually name.
Site identification	site_name	[text]	Site 55 ID	Represents site identification, usually name.
Cell height	height	meters	40	Cell height above the ground.
Cell azimuth	azimuth	degree	50	Cell direction from the north, value ranges from 0 to 360.
Mechanical tilt	tilt	degree	1	Cell mechanical tilt value.
Frequency	frequency	MHz	3500	Frequency value in MHz.
Power	power	dBm	40	Based on Workspace parameter, it can be only Cell power, and EIRP will be calculated from antenna gain and misc loss. It can represent EIRP value too, if Workspace parameter Calculate EIRP is defined to No.
Antenna Gain	Antenna_gain	dBi	18.2	Gain of antenna which is assigned for Cell.
Misc. Loss	Misc_loss	dB	1	Total Cell loss.
Bandwidth	bandwidth	MHz	0.015	Cell bandwidth value in MHz. Especially required for 3G, 4G, and 5G technologies.
Subcarrier spacing	Subcarrier_spacing	kHz	15	Especially required for 5G, as 4G uses constant value 15.
MIMO configuration	tx_mimo	Number	4	Transmitter MIMO configuration, possible values 1, 2, 4, 8, 16, 32, 64.
MIMO configuration	rx_mimo	Number	4	Receiver MIMO configuration, possible values 1, 2, 4, 8, 16, 32, 64.
Cell load	cell_load	Percent	30	Parameter ranges are from 0 to 100 percent. Describes how the cell is loaded in real-time. Load is taken for broadband calculations.
Technology	technology	Text	2G	Possible values: 2G, 3G, 4G, 5G. Describes cell technology.
Antenna name	antenna_id	Number	1	Represents Antenna ID value.

RF Predictions structure

- Predictions
- Results
- Temp



Description: C:\CE_Course\0. Descriptions

Name: 4. Cell Prediction.pdf



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Thank you!

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